

Tutorial Exercises – Series 1

Exercise 1 : Flowcharts

- Draw a flowchart to read the two sides of a rectangle and calculate its area.
- Draw a flowchart to add two numbers entered by the user.
- Draw a flowchart to read a variable $\text{num1} = 7$ and store the negative value of num1 in another variable num2.
- Draw a flowchart to compute the area of a circle of Radius R.
- Draw a flowchart to read a number and identify either the number is positive or negative.
- Draw a flowchart to find all the roots of a quadratic equation $ax^2+bx+c=0$.

Exercise 2 : Arithmetic

Evaluate the following numerical expressions given : $A=6$, $B=7$, $C=4$ and $D=2$

$S1=C*B/D+2*A$	$S5=C*(B/(D+(2*A)))$
$S2=C*(B/D)+2*A$	$S6=((C*B)/D) + (2*A)$
$S3=C*B/(D+2*A)$	$S7=A+B - C/D$
$S4=C*(B/(D+2)*A)$	

Logical expressions/operations

Exercise 3 :

Given that $X, Y, Z, P1, P2, \dots, P7$ are Boolean variables, and A and B are integers, evaluate the following logical expressions under the assignments: $X \leftarrow \text{true}$, $Y \leftarrow \text{not}(X)$, $Z \leftarrow \text{not}(Y)$, $A \leftarrow 20$, and $B \leftarrow 2$.

$P1 \leftarrow X \text{ OR } Y \text{ AND } Z$	$P5 \leftarrow X \text{ AND } Y \text{ OR NOT}(Z)$
$P2 \leftarrow X \text{ OR } Y \text{ AND } Z$	$P6 \leftarrow (A < 4 \text{ OU } P3) \text{ ET } (B + A = 22)$
$P3 \leftarrow (X \text{ OR } Y) \text{ AND } Z$	$P7 \leftarrow (A * B + 5) > 45 \text{ AND NOT } P2$
$P4 \leftarrow X \text{ AND } Y \text{ AND } Z$	

Exercise 4

Let a, b , and c be three Boolean variables. Determine which pairs of the following Boolean expressions are equivalent. For those that are not equivalent, provide an example in which one expression evaluates to true while the other evaluates to false.

1. NOT ($a \text{ OR } b$) with $a \text{ OR } b$	5. $a \text{ AND } (b \text{ OR } c)$ with $a \text{ AND } b \text{ OR } c$
2. NOT($a \text{ AND } b$) with NOT $a \text{ OR NOT } b$	6. $a \text{ AND } (b \text{ OR } c)$ with $(c \text{ OR } b) \text{ AND } a$
3. NOT($a \text{ OR NOT } b$) with NOT $a \text{ AND } b$	7. $a \text{ AND } (b \text{ OR } c)$ with $a \text{ AND } b \text{ OR } a \text{ AND } c$
4. NOT NOT NOT a with NOT a	8. $a \text{ OR } (b \text{ AND } c)$ with $a \text{ OR } b \text{ AND } a \text{ OR } c$

Exercise 5

Consider the following algorithm for solving quadratic equations:

Algorithm Equation2nd

Var

a,b,c,d, sd: real;
cond0, cond1, cond2, cond3, cond4, cond5, cond6, cond7, cond8, cond9 : boolean

Begin

Write ("Enter a, b and c : ") ;

Read (a,b,c)

$d \leftarrow b*b - 4*a*c$

$cond0 \leftarrow (a = 0)$

$cond1 \leftarrow (cond0 \text{ AND } b = 0)$

$cond2 \leftarrow (cond1 \text{ AND } c = 0)$

$cond3 \leftarrow (cond1 \text{ AND } c \neq 0)$

$cond4 \leftarrow (cond0 \text{ AND } b \neq 0)$

$cond5 \leftarrow (a \neq 0)$

$cond6 \leftarrow (cond5 \text{ AND } d < 0)$

$cond7 \leftarrow (cond5 \text{ AND } d \geq 0)$

$cond8 \leftarrow (cond7 \text{ AND } d = 0)$

$cond9 \leftarrow (cond7 \text{ AND } d > 0)$

write("the equation is : ",a,"x^2+ ",b,"x+", c," = 0")

if (.....) then

write("Every number is a solution")

endif

if (.....) then

write("There's no solution")

endif

if (.....) then

write ("x = ", -c/b)

endif

if (.....) then

write("No real number is a solution")

endif

```
if (.....) then
write("x = ", -b/(2*a))
endif
```

```
if (.....) then
```

```
begin
```

```
sd ← sqrt(d)
write("x1 = " , (-b + sd)/(2*a))
write("x2 = " , (-b - sd)/(2*a))
```

```
end if
```

```
END
```

Complete the “if... then” conditions with the appropriate Boolean variables.

Instructions, operations & conditions

Exercise 6

Run the following algorithm :

Algorithm Test1

Var

a, b, c : integer

Begin

```
a ← 15
b ← 51
c ← a
a ← b
b ← c
write (a, b) ;
```

End

- What does this algorithm do?
- Given three variables a, b and c, give an algorithm to permute them circularly, transferring the initial values from a to b, from b to c, and from c to a.

Exercise 7

- Write an algorithm that displays the season by entering the month number.
- Write an algorithm that reads an integer and then displays whether it is even or odd.
- Write an algorithm that reads two integer variables, a and b, and then indicates whether a is a multiple of b, whether b is a multiple of a, or if there is no multiplicative relationship between a and b.

Exercise 8

A year is said to be a leap year (it has 366 days) if it is greater than 1584, and:

- the year is divisible by 4 and not divisible by 100
- or the year is divisible by 400.

Otherwise, the year is not a leap year (it has 365 days).

Write an algorithm that reads a year > 1584 and then indicates whether it is a leap year.

Exercise 9

- Write an algorithm that calculates the square of the maximum of two integers provided by the user.
- Write an algorithm that sorts two integers a, and b provided by the user in ascending order.

Propose three solutions:

- Using only if...then conditions
- Using if...then conditions and the AND operator
- Using if...else conditions and the AND operator

The result display should follow the example below: $A = 5, b = 3$

The order of the variables is as follows: $b < a$

For homework, do the same exercise with three different integers.

Exercise 10

A reprography shop charges 3 dinars for the first ten photocopies, 2 dinars for the next twenty, and 1 dinar for each additional photocopy. Write an algorithm that asks the user for the number of photocopies made and then displays the total cost.